

THASHMIKA RATHNAYAKE

Software Engineer Intern

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Computer Science

PROFESSIONAL SUMMARY

Third year Computer Science undergraduate with full-stack experience in React, NestJS, and MongoDB. Built and deployed core parts of MediGO, a real-time ambulance dispatch platform live at a partner hospital. Including high level encryption and a concurrency fix that closed a real double assignment bug in production. Seeking a Software Engineer Internship to keep building backend systems that handle real operational load.

TECHNICAL SKILLS

- **Languages:** Java, JavaScript, Python, TypeScript, HTML5/CSS3, SQL
- **Backend & APIs:** Node.js (Express.js, NestJS), RESTful APIs
- **Frontend Development:** React.js, HTML5, CSS3, Tailwind CSS
- **Databases:** MySQL, PostgreSQL, MongoDB
- **Tools & Security:** Git, GitHub, Postman

PROJECTS

MediGO: Real-time Ambulance Dispatch & Fleet Management Platform - [MediGO-GitHub](#)

Role: Full-Stack Developer (Team of 5)

Technologies: React, Node.js (NestJS), TypeScript, Google Maps API, Firebase (RTDB/Transactions), Flutter

- **Problem & Architecture:** Built the hospital facing web platform of a three tier emergency coordination system that removes communication delays between hospital dispatchers and ambulance drivers during inter-hospital patient transfers. As one of two full-stack developers on a five member team, owned the React administrative dashboard and a substantial share of the NestJS backend, which synchronises dispatch state, driver location and trip status with a Flutter driver application through a shared Firebase Realtime Database layer, keeping web and mobile clients consistent without polling.
- **Technical Implementation:**
 - Developed 14 of the system's 25 REST endpoints, covering the full transfer request lifecycle - creation, driver assignment, status transitions, and completion.
 - Applied **AES-256 encryption** to sensitive patient fields, consistent across web and mobile clients.
 - Diagnosed and fixed a live concurrency bug where simultaneous driver assignments sent duplicate dispatch requests to the same driver, resolved with **Firebase atomic transactions** to guarantee only one assignment can succeed.
 - Implemented Role Based Access Control across four user roles, combining NestJS route guards with Firebase security rules for role and hospital scoped data access.
 - Built an ETA engine that reacts to live location updates and queries the Google Directions API for real-time arrival estimates.
 - Delivered the dashboard, live ambulance map, fleet management, and transfer management interfaces end to end.
- **Impact & Results:** Deployed the platform and demonstrated it to a partner hospital, which received it positively as it had no comparable dispatch capability in place. Validated ETA accuracy against real arrival times during field testing, with deviations remaining within an acceptable operational margin.

Mini Service Request Board: Service request marketplace | Full-stack Project [MSRB-GitHub](#)

Technologies: React (Next.js), Node.js (Express.js), MongoDB Atlas, Mongoose, Tailwind CSS, Vercel, Railway

- **Overview:** Built a production ready service request marketplace using React (Next.js) and Node.js (Express.js), enabling homeowners to post service requests and tradespeople to browse, filter, and manage jobs in real time.
- **RESTful API Design:** Architected and implemented a clean five endpoint REST API using Node.js and Express.js, following industry standard HTTP semantics (GET, POST, PATCH, DELETE) with proper status codes and a global error handling middleware for consistent, secure error responses.
- **Data Modelling & Validation:** Designed a Mongoose schema with multilayer input validation enforcing required fields, enum constraints, and regex based email format checks at the database layer, paired with client side validation on the frontend to prevent unnecessary API calls.
- **Version Control:** Used **Git and GitHub** for source control and collaborative development throughout the project lifecycle.

EDUCATION

Informatics Institute of Technology (IIT), Sri Lanka (University of Westminster, UK) 2024 – 2028 (expected)
BSc (Hons) Computer Science

- **Current standing: First Class (Current average: 85.7%)**

Nalanda College - Colombo 10

2014 - 2022

- A/L examination - Biological Science Stream (BIO)

EXTRACURRICULAR ACTIVITIES & ACHIEVEMENTS

Hult Prize On-Campus Qualifier – IIT (2026)

- **Competitions & Pitching:** Pitched MediGo at the **Hult Prize On-Campus Qualifier at IIT**, earning a Certificate of Participation for innovative social tech solutions.

Deco Team Volunteer, "Dil Se" Cultural Event - Rotaract Club of IIT (2025)

- Designed and set up venue decorations, collaborating with logistics teams for event execution.

Student Member, IEEE Computer Society Student Branch Chapter of IIT (2024 - 2025)

- Engaged in technical workshops, industry talks, and peer learning sessions.

Nalanda College

- **Handball: All-Island Runners-Up** (U-19, 2018 & U-16, 2016) | **4th Place National Level** (U-20, 2017)
- **Hockey: Gold Star Champions** (U-16, 2016)

SKILLS

- **Soft skills:** Time management, Problem-Solving, Team Collaboration, Leadership, Quick learning
- **Languages:** English, Sinhala

CERTIFICATES

- Python Essential Training - LinkedIn
- Python Data Analysis - LinkedIn

REFEREE

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